

# Robust test statistics

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September 2026

## Abstract

In this note, we provide three robust test statistics which are based on median and median absolute deviation (MAD) estimators, Hodges-Lehmann and Shamos estimators, and Huber location and RC estimators.

## 1 Introduction

One of the widely-used conventional methods to test  $H_0 : \mu = \mu_0$  is to use the  $z$ -test statistic (with  $n \geq 30$ ),

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}},$$

where  $\bar{X}$  is the sample mean and  $S$  is the sample standard deviation. However, this method is very sensitive to data contamination. In this note, we provide three robustified  $z$ -test statistics that can be used when the sample size is large. However, when the sample size is small, it may not be appropriate to use the standard normal distribution. In this note, we provide a method for obtaining the empirical distributions of these test statistics. These results are implemented in the `rt.test` package.

## 2 Robustified test statistics

For the sake of completeness, we briefly review the test statistics proposed by [1], [2], and [3]. By replacing the mean and the standard deviation with robust location and scale estimators, they proposed the following robustified  $z$ -test statistics. These are given by

$$T_A = \sqrt{\frac{2n}{\pi}} \Phi^{-1}\left(\frac{3}{4}\right) \cdot \frac{\text{median}_{1 \leq i \leq n} X_i - \mu}{\text{median}_{1 \leq i \leq n} |X_i - \text{median}_{1 \leq i \leq n} X_i|} \xrightarrow{d} N(0, 1), \quad (1)$$

$$T_B = \sqrt{\frac{6n}{\pi}} \Phi^{-1}\left(\frac{3}{4}\right) \frac{\text{median}_{i \leq j} \left( \frac{X_i + X_j}{2} \right) - \mu}{\text{median}_{i \leq j} (|X_i - X_j|)} \xrightarrow{d} N(0, 1), \quad (2)$$

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Table 1: Breakdown Points and AREs of the estimators under consideration.

|           | Location |        |     | SD   | Scale |     |        |
|-----------|----------|--------|-----|------|-------|-----|--------|
|           | Mean     | Median | HL  |      | IQR   | MAD | Shamos |
| Breakdown | 0%       | 50%    | 29% | 0%   | 25%   | 50% | 29%    |
| ARE       | 100%     | 64%    | 96% | 100% | 38%   | 37% | 86%    |

where  $\Phi^{-1}(\cdot)$  is the inverse of the standard normal cumulative distribution function and  $\xrightarrow{d}$  denotes convergence in distribution.

### 3 Empirical distributions

As mentioned above, the robustified statistics,  $T_A$  in (1) and  $T_B$  in (2), converge to the standard normal distribution. However, when the sample size is small, it is not appropriate to use the standard normal distribution.

It may be impossible to obtain the theoretical distributions of  $T_A$  and  $T_B$ . Thus, we obtain the empirical distributions of  $T_A$  and  $T_B$  using extensive Monte Carlo simulations and calculate their empirical quantiles, which are useful for estimating critical values, confidence intervals,  $p$ -values, etc. We used the R language to conduct the simulations, which are summarized as follows. We generated one hundred million ( $N = 10^8$ ) samples of size  $n$  from the standard normal distribution to obtain the empirical distributions of  $T_A$  and  $T_B$ , where  $n = 4, 5, \dots, 50$ . Using these samples, we can obtain the empirical distribution of  $T_A$  or  $T_B$  for each sample size  $n$ . Then, by inverting the empirical distribution, we obtained the empirical quantiles of  $p$ .

It is worthwhile to discuss the accuracy of the empirical quantiles obtained above. Let  $F_N^{-1}(p)$  be the empirical quantile of  $p$  obtained from the  $N$  replications and  $F^{-1}(p)$  be the true quantile of  $p$ . Then it is easily seen from Corollary 21.5 of [4] that the sequence  $\sqrt{N}(F_N^{-1} - F^{-1}(p))$  is asymptotically normal with mean zero and variance  $p/(1-p)/f^2(F^{-1}(p))$ . Thus, the standard deviation of the empirical quantile of  $p$  is approximately proportional to  $\sqrt{p(1-p)/N}$ , which has its maximum value at  $p = 0.5$ . Consequently, the empirical quantile values are computed with an approximate accuracy of  $0.5/\sqrt{N}$ . With  $N = 10^8$ , we have  $0.5/\sqrt{N} = 0.00005$ , which roughly indicates that the empirical quantile values are accurate up to the fourth decimal point.

Given that the probability density functions of  $T_A$  and  $T_B$  are symmetric about zero, we have  $F(-x) = 1 - F(x)$  and  $F^{-1}(1/2) = 0$ . Letting  $q_p$  be the  $p$ th lower quantile so that  $F(q_p) = p$ , we have  $q_{1-p} = -q_p$ . Thus, it is enough to find the  $p$ th quantile only when  $p > 1/2$ . Let  $G(\cdot)$  be the cumulative distribution function of  $|X|$ . Then we have

$$G(x) = P[|X| \leq x] = P[-x \leq X \leq x] = F(x) - F(-x) = 2F(x) - 1.$$

Substituting  $x = q_p$  into the above, we have  $G(q_p) = 2p - 1$ . Thus, we have

$$q_p = G^{-1}(2p - 1),$$

which is more efficient than using  $q_p = F^{-1}(p)$  when obtaining empirical quantile values.

## References

- [1] C. Park. Note on the robustification of the Student  $t$ -test statistic using the median and the median absolute deviation. <https://arxiv.org/abs/1805.12256>, 2018. ArXiv e-prints.
- [2] R. Jeong, S. B. Son, H. J. Lee, and H. Kim. On the robustification of the  $z$ -test statistic. Presented at KIIE Conference, Gyeongju, Korea. April 6, 2018, 2018.
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- [4] A. W. van der Vaart. *Asymptotic statistics*. Cambridge University Press, Cambridge, 1998.